

Class 7 General Science — SA-1 Examination Question Paper

Summative Assessment - I (2026-27) | Maximum Marks: 80 | Time: 2 Hours 30 Minutes

Class 7 General Science — Summative Assessment (SA-1) Official Question Paper

Academic Year 2026-27 | Maximum Marks: 80 | Time: 2 Hours 30 Minutes

Textbook: Science Trek - 7 | Syllabus: Chapters 5 & 6

General Instructions:

1. All questions are compulsory.
2. Section A (20 Marks): Multiple Choice Questions & Objective Fill in the Blanks.
3. Section B (15 Marks): True/False, Analogies & Match the Following.
4. Section C (15 Marks): Picture-Based Questions (Experiments & Observational Analysis).
5. Section D (15 Marks): Short & Long Descriptive Analytical Questions.
6. Section E (15 Marks): Case-Based Competency & Higher Order Thinking Skills (HOTS).

SECTION A: Objective & Conceptual Multiple Choice (20 Marks)

Part I: Choose the Correct Option ($10 \times 1 = 10$ Marks)

1. Which of these is a physical change?
 - (a) Cooking food
 - (b) Rusting of iron
 - (c) Boiling water
 - (d) Burning wood
2. Which of the following occurs during a chemical change?
 - (a) Melting of ice
 - (b) Breaking of glass
 - (c) Formation of a new substance
 - (d) Tearing paper
3. What is the product of combustion?
 - (a) Water
 - (b) Ash and smoke
 - (c) Oxygen
 - (d) Ice

4. Which process is an example of crystallisation?
 - (a) Freezing of water
 - (b) Boiling of water
 - (c) Formation of salt crystals from seawater
 - (d) Cutting paper
 5. What happens when iron rusts?
 - (a) It turns into water
 - (b) It turns into a new substance
 - (c) It melts
 - (d) It changes shape
 6. Adolescence usually occurs in the age group of:
 - (a) 0–5 years
 - (b) 8–10 years
 - (c) 11–19 years
 - (d) 20–25 years
 7. Which hormone is mainly responsible for the production of sperms and male secondary characteristics?
 - (a) Oestrogen
 - (b) Testosterone
 - (c) Adrenaline
 - (d) Thyroxine
 8. The menstrual cycle usually occurs once every:
 - (a) 7–10 days
 - (b) 15–20 days
 - (c) 28–30 days
 - (d) 45–60 days
 9. Hormones are produced by special organs called:
 - (a) Veins
 - (b) Bones
 - (c) Glands
 - (d) Muscles
 10. Excess oil on the face during adolescence causes:
 - (a) Freckles
 - (b) Wrinkles
 - (c) Acne
 - (d) Scars
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Part II: Fill in the Blanks (10 × 1 = 10 Marks)

1. ____ occurs when water evaporates, leaving behind solid crystals.
2. A chemical process in which a substance reacts with oxygen with evolution of heat is called ____.
3. ____ is a physical change where the shape of an object changes but its material remains the same.
4. The process in which iron reacts with oxygen and moisture is called _ **and this is a** _ change.
5. Burning of magnesium is a _ **change in nature, producing a substance called** _.

6. The period of rapid growth and change in the body between childhood and adulthood is called ____.
7. Chemical substances that control physical and emotional changes during adolescence are called ____.
8. The hormone ____ is responsible for male secondary sexual characteristics.
9. ____ foods provide energy for daily work and play.
10. ____ is the government campaign promoting a drug-free India.

SECTION B: True/False, Analogies & Match the Following (15 Marks)

Part I: State True or False ($5 \times 1 = 5$ Marks)

1. Chemical changes cannot be reversed to restore the original substance.
2. Combustion requires fuel, oxygen and heat to sustain the process.
3. Boiling water is a physical change, not a chemical one.
4. Crystallisation is a reversible change where dissolved substances form crystals.
5. Rusting of iron is an irreversible change that forms rust.

Part II: Analogies ($5 \times 1 = 5$ Marks)

1. Burning wood : Ash :: Iron : ____
2. Boiling water : Steam :: Melting ice : ____
3. Crystallisation : Physical change :: Combustion : ____
4. Testosterone : Boys :: Oestrogen : ____
5. Childhood : Slow growth :: Adolescence : ____

Part III: Match the Following ($5 \times 1 = 5$ Marks)

1. Match Column A with Column B:

| Column A | Column B |

| :--- | :--- |

| (1) Hormones | (a) Deep voice and facial hair |

| (2) Adolescence | (b) Online harassment |

| (3) Oestrogen | (c) Produced by glands |

| (4) Testosterone | (d) Menstrual cycle in girls |

| (5) Cyberbullying | (e) Transition to adulthood |

SECTION C: Picture-Based Experimental Analysis (15 Marks)

Question 32: Crystallisation Process (5 Marks)

Observe an evaporating dish heated on a wire gauze with blue vitriol (copper sulphate) solution:

- (a) Which separation process is demonstrated when dissolved solid forms geometric shapes upon cooling? (2 Marks)
- (b) Is this process a physical or chemical change? Justify your answer. (3 Marks)

Question 33: Lime Water Test Experiments (5 Marks)

Four test tube setups (a, b, c, d) were connected via delivery tubes to lime water:

- Setup (a): Vinegar + Baking Soda
- Setup (b): Lemon Juice + Vinegar
- Setup (c): Vinegar + Common Salt
- Setup (d): Lemon Juice + Baking Soda

- (a) In which setup(s) does lime water turn milky? (2 Marks)
- (b) Write the chemical name and formula of the gas evolved that causes lime water to turn milky. (1 Mark)
- (c) Write the chemical equation explaining why the lime water becomes milky. (2 Marks)

Question 34: Growth Spurt & Nutrition in Adolescence (5 Marks)

Observe a diagram illustrating adolescent students of the same age showing noticeable height variations and food choices:

- (a) What biological phenomenon explains varying rates of height increase during puberty? (2 Marks)
- (b) Identify two critical factors that govern maximum height potential. (1 Mark)
- (c) Why should adolescents choose fresh pulses, milk, and vegetables over junk food? (2 Marks)

SECTION D: Short & Long Descriptive Questions (15 Marks)

1. Answer in 1–2 sentences: ($3 \times 1 = 3$ Marks)

- (a) How does oxygen contribute to combustion?
- (b) Name two hormones that play a role in adolescent changes.
- (c) What is the primary function of Vitamin B12 in the body?

2. Short Answer Questions ($3 \times 2 = 6$ Marks):

- (a) How do physical changes differ from chemical changes? Give one example each.
- (b) What are the observable signs indicating that a chemical change has taken place?
- (c) Why is it essential for adolescent girls to maintain menstrual hygiene?

3. Long Answer / Categorisation Question ($1 \times 6 = 6$ Marks):

- (a) How does rusting weaken iron, and what are three distinct methods to prevent it? (3 Marks)
- (b) Categorise the following changes into: (A) Physical changes only, (B) Chemical changes only, (C) Both physical and chemical changes:

Tearing of paper, Burning of a candle, Rusting of iron, Melting of ice, Curdling of milk, Burning of magnesium ribbon. (3 Marks)

SECTION E: Competency-Based Case Study & HOTS (15 Marks)

Question 38: Case Study 1 — Decomposition & Weathering (5 Marks)

A wooden log was left outside in the rain and gradually began to rot as fungi and bacteria decomposed the organic material.

- (a) Identify the type of change (physical or chemical) occurring in the decaying log. Explain why. (2 Marks)
- (b) Natural agents like freezing water in rock crevices crack rocks into soil. Is soil formation purely a physical change or does it involve chemical weathering? Explain. (3 Marks)

Question 39: Assertion & Reason Analysis (4 Marks)

Choose: (A) Both A and R true, R correct explanation; (B) Both true, R not correct explanation; (C) A true, R false; (D) Both false.

(a) **Assertion (A):** Rusting of iron is a chemical change.

Reason (R): A new substance is formed when iron reacts with oxygen and water. (2 Marks)

(b) **Assertion (A):** Adolescence is a phase of rapid physical growth and change.

Reason (R): Hormones remain inactive during adolescence, affecting physical growth. (2 Marks)

Question 40: Higher Order Thinking Skills (HOTS) (6 Marks)

(a) Mary and Manoj are classmates. Mary developed a little bulge on the front of her neck and her doctor prescribed an iodine-rich diet. Manoj developed a visible bump on his throat, but the doctor assured him it was entirely normal for boys his age. What scientific reasons account for the different medical advice? [NCERT] (3 Marks)

(b) When a magnesium ribbon is burned in air:

(i) Write the balanced chemical equation.

(ii) Explain why this combustion is a chemical change, stating the three requirements of the Fire Triangle. (3 Marks)